

Housing Reallocation and Demographic Transitions

A two-generation model

Environment

Agents. Time is discrete and one period is one generation. At date t , the economy contains Y_t young households and O_t old households. A young household has income y_i and liquid wealth b_i , drawn from a fixed type distribution F . A young renter becomes an old renter, a young owner becomes an old incumbent, and every old household exits at the end of the period.

Preferences. A young household values adult consumption c , space net of children's needs, and completed fertility:

$$u_t^Y(c, h, n) = \log c + \alpha \log(h - \kappa n) + \vartheta_t \log n. \quad (1)$$

Each child uses $\chi > 0$ units of goods and $\kappa > 0$ units of space. An old household values consumption, retained housing, and the estate delivered at death:

$$u^O(c^O, h^O, e) = \log c^O + \gamma \log h^O + \omega_B \log e. \quad (2)$$

The financial sector supplies liquid claims and absorbs estates at price q .

Housing and financial markets. A fixed stock $\bar{H}$ produces one unit of housing services per unit and period. Renters can occupy at most $\bar{h}^R$, while owners can occupy at most $\bar{h}^O > \bar{h}^R$. The asset price is P_t , one-period-ahead goods have price $q \in (0, 1)$, and $R = q^{-1}$. Competitive housing intermediaries imply the user-cost relation:

$$r_t = (1 + \tau_t^p)P_t - qP_{t+1}, \quad (3)$$

where $\tau_t^p P_t$ is the recurring property tax per unit.

Ownership and taxation. A buyer finances the share ϕ of the purchase with a one-period mortgage and posts $(1 - \phi)P_t h$ at closing. The mortgage advance is $\phi P_t h$ and its next-period repayment is $R\phi P_t h$. Selling a unit realizes the capital-gains tax

$$\ell_t = \tau^g [P_t - P_{t-1}]_+, \quad (4)$$

while a unit retained until death enters the estate with stepped-up basis. The tax is zero whenever the real asset price is constant. These cash flows give two tenure-specific costs. Buying a unit, paying its property tax, and selling it next period costs:

$$r_t^O = (1 + \tau_t^p)P_t - q(P_{t+1} - \ell_{t+1}) = r_t + q\ell_{t+1}. \quad (5)$$

For an old incumbent, retaining a unit rather than selling it, paying the property tax, and placing its next-period value in the estate costs:

$$r_t^I = (P_t - \ell_t) + \tau_t^p P_t - qP_{t+1} = r_t - \ell_t. \quad (6)$$

Thus the buyer anticipates the future tax, while the incumbent avoids the current tax by retaining the house.

Demography. Each child generates $\nu > 0$ future young households after survival and household formation. The children of a household affect future cohort size but do not inherit that household's individual state.

Household problems

Old households

An old incumbent enters with net financial wealth a and a house H . In current-value terms, selling the entire house produces $(P_t - \ell_t)H$; retained housing costs $r_t - \ell_t$ by equation (6). Her problem is therefore:

$$\begin{aligned} V_t^O(a, H) &= \max_{c^O, h^O, e} u^O(c^O, h^O, e) \\ \text{subject to } &c^O + qe + (r_t - \ell_t)h^O = a + (P_t - \ell_t)H + T_t, \quad 0 < h^O \leq H, \quad e \geq P_{t+1}h^O. \end{aligned} \quad (7)$$

The estate includes the next-period value of retained housing. When its composition constraint and $h^O \leq H$ are slack, the incumbent's marginal value of retained space is:

$$\frac{u_h^O}{u_c^O} = \frac{\gamma c_t^O}{h_t^O} = r_t - \ell_t. \quad (8)$$

An old renter has no house or gain and solves the same problem with budget $c^O + qe + r_t h^O = a + T_t$.

Young households

Conditional on renting, a young household chooses current consumption, housing, fertility, and saving:

$$W_t^R(i) = \max_{c,h,n,a'} u_t^Y(c, h, n) + \beta V_{t+1}^R(Ra') \quad (9)$$

subject to $c + \chi n + a' + r_t h = y_i + b_i + T_t, \quad 0 < h \leq \bar{h}^R, \quad a' \geq 0.$

Conditional on owning, the same household purchases h units, posts the down payment at the beginning of the period, and repays the mortgage when old:

$$W_t^O(i) = \max_{c,h,n,a'} u_t^Y(c, h, n) + \beta V_{t+1}^O(a_{t+1}, h) \quad (10)$$

subject to $c + \chi n + a' + [(1 - \phi) + \tau_t^p] P_t h = y_i + b_i + T_t,$
 $a_{t+1} = Ra' - R\phi P_t h, \quad (1 - \phi) P_t h \leq b_i,$
 $0 < h \leq \bar{h}^O, \quad a' \geq 0.$

Income received after closing can finance consumption and saving but cannot be pledged for the down payment. The household chooses the tenure with the higher value.

Suppose saving is interior and the future old retention limit is slack. Let λ_t^m be the multiplier on the young budget, μ_t the multiplier on the owner's down-payment constraint, and η_t^m the multiplier on the tenure-specific size limit. The access values in current goods are:

$$\zeta_t^R = \frac{\eta_t^R}{\lambda_t^R}, \quad \zeta_t^O = \underbrace{\frac{\mu_t}{\lambda_t^O} (1 - \phi) P_t}_{\zeta_t^{O,F}} + \underbrace{\frac{\eta_t^O}{\lambda_t^O}}_{\zeta_t^{O,S}}. \quad (11)$$

Using the saving Euler equation and the old value-function envelopes, the housing conditions reduce to:

$$\frac{\alpha c_t^m}{h_t^m - \kappa n_t^m} = r_t^m + \zeta_t^m, \quad r_t^R = r_t, \quad r_t^O = r_t + q\ell_{t+1}. \quad (12)$$

The ownership expression follows from the purchase, mortgage, property-tax, and resale cash flows in equation (10). The fertility condition prices the goods and space required by a marginal child:

$$\frac{\vartheta_t c_t^m}{n_t^m} = \chi + \kappa(r_t^m + \zeta_t^m). \quad (13)$$

A housing constraint raises the household's shadow price of a child by $\kappa\zeta_t^m$.

Steady states and transition

Let $\bar{n}_t$, $\bar{h}_t^Y$, and $\bar{h}_t^O$ denote fertility and housing averaged over the young type distribution and the inherited old distribution. Let $\bar{s}_t^O$ denote housing sold by an average old incumbent. Cohort sizes and the housing stock obey:

$$Y_{t+1} = \nu \bar{n}_t Y_t, \quad O_{t+1} = Y_t, \quad Y_t \bar{h}_t^Y + O_t \bar{h}_t^O = \bar{H}. \quad (14)$$

Fertility determines the next young cohort, while current young and old households compete for the fixed housing stock.

The government rebates property- and capital-gains-tax receipts equally across living households:

$$(Y_t + O_t) T_t = \tau_t^p P_t \bar{H} + \ell_t O_t \bar{s}_t^O. \quad (15)$$

The taxes remain private wedges, while their revenue stays within the household sector.

For a constant fertility preference ϑ , let $\mathcal{N}(P, T; \vartheta)$, $\mathcal{H}^Y(P, T; \vartheta)$, and $\mathcal{H}^O(P, T; \vartheta)$ denote stationary fertility and housing choices aggregated across household types.

Definition 1 (Positive steady state). A positive steady state is a constant equilibrium with $Y^* = O^* > 0$, a constant asset price and transfer, and stationary household policies and distributions. Its aggregate conditions are:

$$1 = \nu \mathcal{N}(P^*, T^*; \vartheta), \quad (16)$$

$$\bar{H} = Y^* [\mathcal{H}^Y(P^*, T^*; \vartheta) + \mathcal{H}^O(P^*, T^*; \vartheta)], \quad (17)$$

$$2Y^* T^* = \tau^p P^* \bar{H}. \quad (18)$$

The replacement condition determines whether a positive stationary population is feasible; housing clearing and the balanced rebate jointly determine its price, transfer, and scale. Because $P_t = P_{t-1}$, equation (4) gives $\ell^* = 0$ in every steady state.

Consider two fertility preferences, ϑ^0 and ϑ^1 , for which the steady-state conditions admit positive equilibria $\mathcal{E}^0$ and $\mathcal{E}^1$. At date zero the preference changes permanently:

$$\vartheta_t = \begin{cases} \vartheta^0, & t < 0, \\ \vartheta^1, & t \geq 0. \end{cases} \quad (19)$$

The shock changes current choices immediately and future cohort sizes through equation (14).

Definition 2 (Transition equilibrium). Given the initial steady state $\mathcal{E}^0$, a transition equilibrium is a sequence of prices, transfers, household policies, old-household distributions, and cohort masses such that households solve equations (7)–(10); the old distribution is generated by the preceding young choices; housing clears and the government budget balances at every date; equation (3) prices the housing asset; equation (14) advances the cohorts; $P_{-1} = P^0$ and the date-zero old enter with the state inherited from $\mathcal{E}^0$; and the sequence converges to $\mathcal{E}^1$.

Capital gains can arise when the asset price rises along the transition, but they vanish at both stationary endpoints.

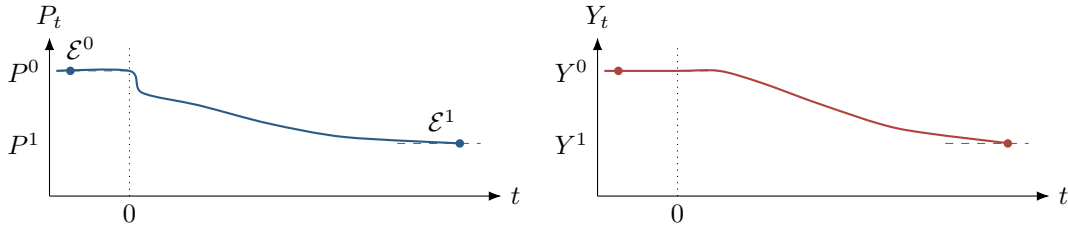


Figure 1: A permanent preference change moves prices immediately, while cohort mass adjusts with a lag through $Y_{t+1} = \nu \bar{n}_t Y_t$. The monotone paths shown are illustrative; a transition equilibrium may adjust nonmonotonically.

Intergenerational housing allocation

Consider an old incumbent at an interior retention choice and a young owner whose down-payment constraint binds while the owner-size limit is slack. Their marginal values of current housing services are:

$$MV_{it}^Y = r_t + q\ell_{t+1} + \zeta_{it}^{O,F}, \quad MV_{jt}^O = r_t - \ell_t. \quad (20)$$

The market component cancels when a unit is transferred between them.

Definition 3 (Conditional housing efficiency). At date t , an allocation is conditionally efficient if no infinitesimal reallocation of the existing housing stock, accompanied by balanced goods transfers among the affected living households and the government, can make all affected households weakly better off and one strictly better off. The comparison holds fertility, cohort masses, tenure assignments, saving choices, and the price sequence fixed.

Proposition 1 (Compensated intergenerational reallocation). *Hold the price sequence, fertility, cohort masses, tenure assignments, and saving choices fixed. Suppose an institution can transfer one marginal unit of existing housing from household j to household i , compensate all affected households and the government, and incurs real cost K_{ijt} per unit. The goods-equivalent surplus is:*

$$\mathcal{S}_{ijt} = \zeta_{it}^{O,F} + q\ell_{t+1} + \ell_t - K_{ijt}. \quad (21)$$

The equilibrium allocation is conditionally inefficient whenever $\mathcal{S}_{ijt} > 0$.

Proof. Increase the young household's housing by dh and reduce the incumbent's retained housing by the same amount. Compensating the two households at their marginal utilities leaves $(MV_{it}^Y - MV_{jt}^O - K_{ijt})dh$ units of goods. Substituting equation (20) gives equation (21). Equation (15) returns the two tax payments to the household sector, so they are private wedges rather than resource costs. $\square$

Corollary 1 (Steady-state misallocation). At either steady state, $\ell_t = \ell_{t+1} = 0$, and the surplus reduces to:

$$\mathcal{S}_{ij}^* = \zeta_i^{O,F} - K_{ij}. \quad (22)$$

A financing-constrained young owner and an old incumbent are conditionally misallocated whenever $\zeta_i^{O,F} > K_{ij}$.

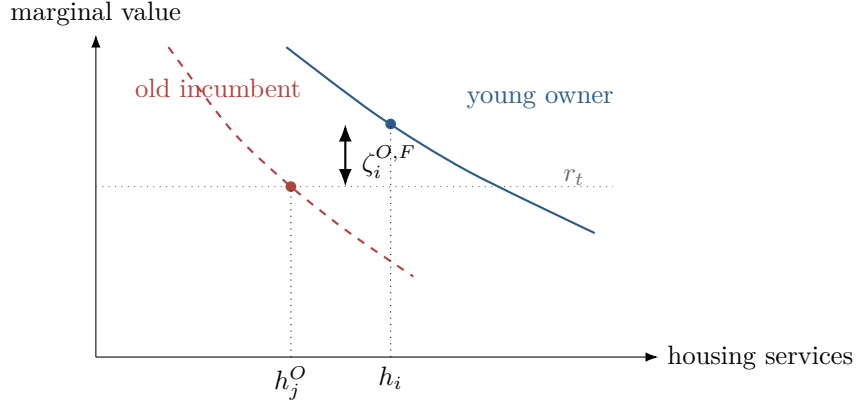


Figure 2: In steady state, a constrained young owner values a marginal unit above the common user cost, while an unconstrained incumbent values it at that cost. The vertical distance is the financing component of the reallocation surplus.

Capital gains therefore amplify the same allocation gap on appreciation dates along the transition; they are absent from its stationary endpoints.

The reallocation result and the fertility response are distinct. Let $A = 1 + \gamma + \omega_B$ and $k = \beta A$. With interior saving and old choices, substituting the old allocation and the saving Euler equation into either young problem gives the lifetime resource identity:

$$(1 + k)c + \chi n + r_t^m h = \mathcal{W}_t, \quad \mathcal{W}_t = y_i + b_i + T_t + qT_{t+1}. \quad (23)$$

Suppose an access constraint fixes the household's housing at $\hat{h}$ while fertility remains interior. Holding $\mathcal{W}_t$ and prices fixed, implicitly differentiating the fertility condition yields:

$$\frac{dn}{d\hat{h}} = \frac{\frac{\alpha\kappa}{(\hat{h} - \kappa n)^2} - \frac{(1+k)\chi r_t^m}{(\mathcal{W}_t - r_t^m \hat{h} - \chi n)^2}}{\frac{\vartheta_t}{n^2} + \frac{(1+k)\chi^2}{(\mathcal{W}_t - r_t^m \hat{h} - \chi n)^2} + \frac{\alpha\kappa^2}{(\hat{h} - \kappa n)^2}}. \quad (24)$$

The first term in the numerator is the value of additional family space; the second is the consumption cost of occupying more housing. Equation (12) implies $[\mathcal{W}_t - r_t^m \hat{h} - \chi n]/[\hat{h} - \kappa n] = (1+k)(r_t^m + \zeta_t^m)/\alpha$. Since $\zeta_t^m \geq 0$, a sufficient condition for greater housing access to raise fertility is:

$$\chi \leq \frac{(1+k)\kappa r_t^m}{\alpha}. \quad (25)$$

Under this condition, an institution that reaches a positive financing gap and increases the household's housing both improves the conditional allocation and raises fertility among the constrained households it directly affects. General-equilibrium prices, rebates, and cohort sizes then propagate that response through equations (14)–(15).